

Joseph Musielewicz

jmusiele@alumni.cmu.edu | jmusiel.github.io

Education

Carnegie Mellon University, PhD in Chemical Engineering – Pittsburgh, PA

Jan 2020 – Feb 2025

- Advised by John Kitchin and Zachary Ulissi.
- Emphasis in Machine Learning.

Iowa State University, BS in Chemical Engineering – Ames, IA

Aug 2015 – Dec 2019

- Graduated summa cum laude.
- Minor in Computer Science.

Professional Experience

Machine Learning Scientist for Chemistry, Entalpic AI – Paris, France

Mar 2025 – present

- Lead end-to-end active learning pipelines for discovering novel materials and molecules, spanning generative sampling, model inference, acquisition, and simulation workflows.
- Achieved 10x enrichment factor in computationally discovered molecules by combining property data mined from scientific papers with simulated molecular features.
- Delivered 100x faster discovery of global optima for a client materials simulation using black-box optimization methods.
- Achieved state-of-the-art fine-tuning performance across domains with 100x smaller datasets.
- Reduced production ML inference costs by over \$100K with efficient dynamic batching.
- Scaled neural network embedding lookups to screen hundreds of millions of candidates using FAISS-backed similarity search.
- Developed reinforcement-learning rewards and uncertainty/diversity acquisition methods for generative materials discovery.
- Mentored junior researchers and coauthored TriForces, a paper accepted at ICML which demonstrates state-of-the-art fine-tuning performance for GNNs on OMat and MatBench tasks.

Research Scientist Intern, Meta AI – San Francisco, CA

May 2023 – Oct 2023

- Benchmarked uncertainty prediction methods for use in the Open Catalyst Demo.
- Developed novel improvements to uncertainty methods for graph neural network potentials resulting in a 50% improvement in uncertainty calibration on the relaxed structure to relaxed energy task.
- Implemented transformer models, conformal prediction, and ensemble methods for estimating GNN uncertainty.
- Presented a manuscript documenting these results at NeurIPS 2023.

Process Development Intern, Genentech – South San Francisco, CA

May 2019 – Aug 2019

- Implemented a computer vision tool to analyze images of a cell culture process and control the bioreactor in real time, saving an estimated 10 operator hours per week.

Ph.D. Research Experience

Ph.D. Researcher, Carnegie Mellon University – Pittsburgh, PA

Nov 2020 – Feb 2025

- Investigated and developed novel active learning, fine tuning, and uncertainty quantification methods for graph neural network models in computational catalysis.
- Collaborated with researchers at Meta AI on the Open Catalyst Project to improve training methods and uncertainty quantification for their machine learned interatomic potentials.
- Developed the Finetuna Python package on GitHub, implementing active learning methods for accelerating atomistic simulations by 90% over competing algorithms.
- Implemented transition state finding methods which converge 3 times more often than previous state of the art.
- Improved fine tuning methods for training machine learned force fields and molecular property prediction.
- Maintained the Ulissi Group Kubernetes cluster for high performance computing and model training.

Personal Projects

auto-goldfish.vercel.app, Auto-Goldfish

2024 – present

- Built Monte Carlo and Hyperband tooling to optimize complex strategy-game decks through parallel simulations.
- Built LLM labeling pipelines with Claude, Gemini, and Ollama using structured schemas and few-shot prompts.

Skills

ML Frameworks: PyTorch, PyTorch Geometric, PyTorch Lightning, Scikit-learn, FAISS, Weights & Biases, Hydra.

Python & Scientific Computing: Python (NumPy, SciPy, Pandas, Pydantic), Jupyter notebooks, Matplotlib. Atomistic simulation libraries including VASP, ASE, and Pymatgen.

Infrastructure & Computing Tools: Git, GitHub Actions, Docker. Distributed HPC and cloud computing using AWS, GCP, Kubernetes, and Slurm. Apache Airflow for workflow orchestration. PostgreSQL, MongoDB, LMDB.

Machine Learning Techniques: Fine-tuning, self-supervised pretraining, reinforcement learning, active learning, uncertainty quantification, conformal prediction, embedding similarity search.

LLMs & AI Agents: LLM API integration (Claude, Gemini, Ollama). Coding agents (Claude Code, Conductor). LLM-powered data extraction from scientific papers. Custom agent skills for Slurm cluster workflows.

Publications

TriForces: Augmenting Atomistic GNNs for Transferable Representations May 2026

A. Ramlaoui, A. Duval, H. Bull, V. Schmidt, H. Talbot, F. D. Malliaros, *J. Musielewicz*
(Accepted to ICML 2026; arXiv:2605.20581)

LeMat-Traj: A Scalable and Unified Dataset of Materials Trajectories for Atomistic Modeling Aug 2025

A. Ramlaoui, M. Siron, I. Djafar, *J. Musielewicz*, A. Rossello, V. Schmidt, A. Duval
(arXiv:2508.20875)

Accessing Numerical Energy Hessians with Graph Neural Network Potentials and Their Application in Heterogeneous Catalysis Feb 2025

J. Musielewicz, B. Wander, R. Cheula, J. Kitchin
(The Journal of Physical Chemistry C)

Improved Uncertainty Estimation of Graph Neural Network Potentials Using Engineered Latent Space Distances Nov 2024

J. Musielewicz, J. Lan, M. Uyttendaele, J. Kitchin
(The Journal of Physical Chemistry C)

Generalization of Graph-Based Active Learning Relaxation Strategies Across Materials Apr 2024

J. Musielewicz, X. Wang, R. Tran, S. Ethirajan, X. Fu, H. Mera, J. Kitchin, R. Kurchin, Z. Ulissi
(Machine Learning: Science & Technology)

Robust and scalable uncertainty estimation with conformal prediction for machine-learned interatomic potentials Dec 2022

Y. Hu, *J. Musielewicz*, Z. Ulissi, A. Medford
(Machine Learning: Science & Technology)

Finetuna: Fine-tuning accelerated molecular simulations Sept 2022

J. Musielewicz, X. Wang, T. Tian, Z. Ulissi
(Machine Learning: Science & Technology)

Leadership Experience

Research Mentor, Entalpic AI – Paris, France Mar 2025 – present

- Mentor a PhD student researcher on graph neural network research focused on self-supervised learning, pretraining, equivariance, and model distillation.
- Coauthored TriForces, an accepted ICML paper that separates composition and structure information to preserve transferable representations and improve GNN fine-tuning on OMat and MatBench tasks.

Process Team Leader, NASA X-Hab Challenge – Ames, IA Aug 2018 – May 2019

- Led a team of undergraduate engineers in an interdisciplinary effort to design a working prototype carbon scrubber which was tested on the Artemis missions.
- Collaborated with researchers at the Ames National Laboratory to model and test a CO₂ adsorbing bed incorporating novel metal organic frameworks.

Nationals Team Leader, ISU Chem-E-Car – Ames, IA June 2017 – Nov 2018

- Led a multidisciplinary team of other engineering students in setting the record for Iowa State at the national Chem-E-Car competition at AIChE 2018.